13

Electromagnetic Induction

13.4

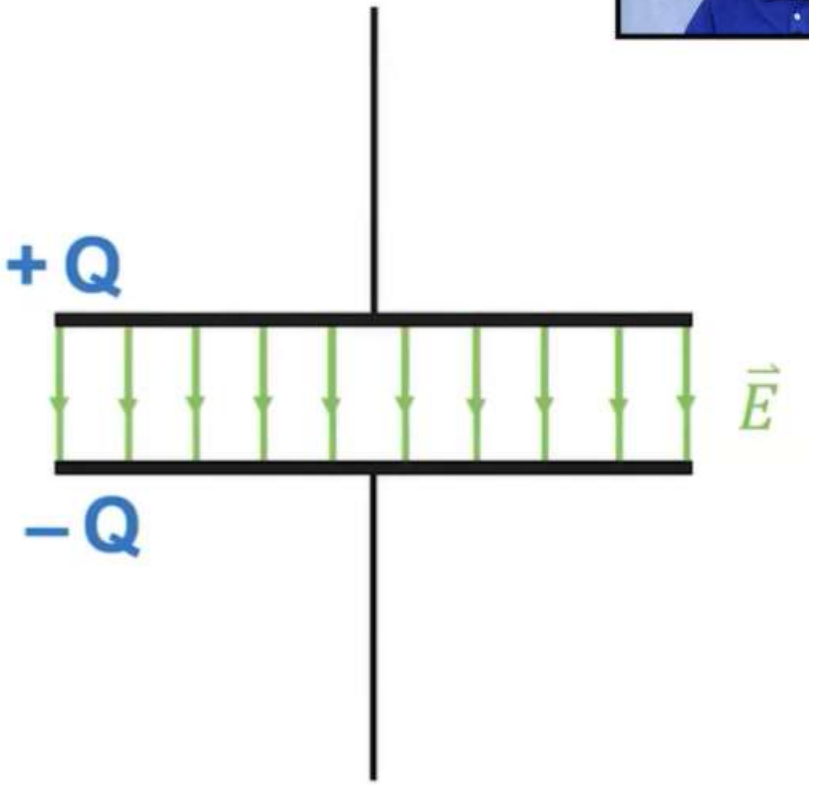
Inductance

13.4.1

Inductance, inductors, and the inductance of a solenoid

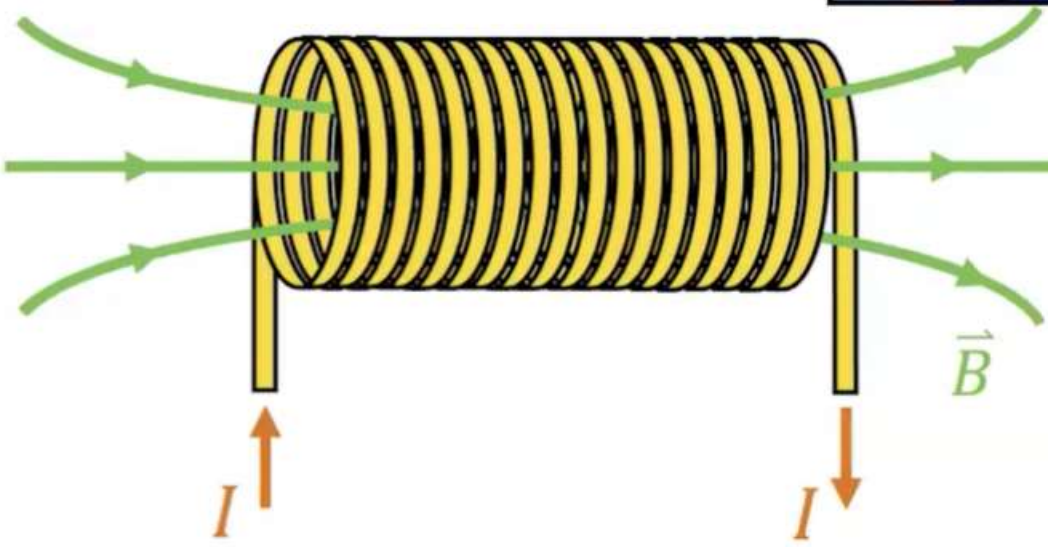
The Parallel Plate Capacitor

- The parallel plate capacitor, when charged, holds equal amounts of positive and negative charges on either plate.
- A constant electric field existed between the plates.
- Based on this physical construction, the capacitor is a device that can store energy in its electric field.



The Solenoid

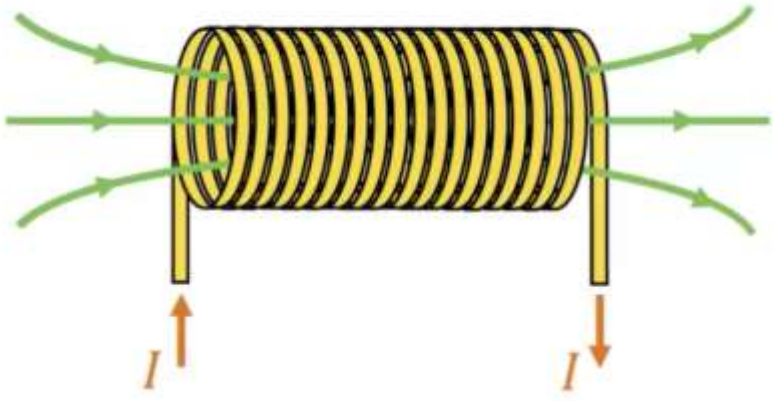
- The solenoid consists of a wire tightly wrapped around a cylindrical core to create a helical stack of conducting loops.
- When a current is established in the solenoid, a magnetic field exists through the core.
- The magnetic field strength inside the solenoid can be determined by the equation



$$B_{sol} = \mu_0 \frac{N}{l} I$$

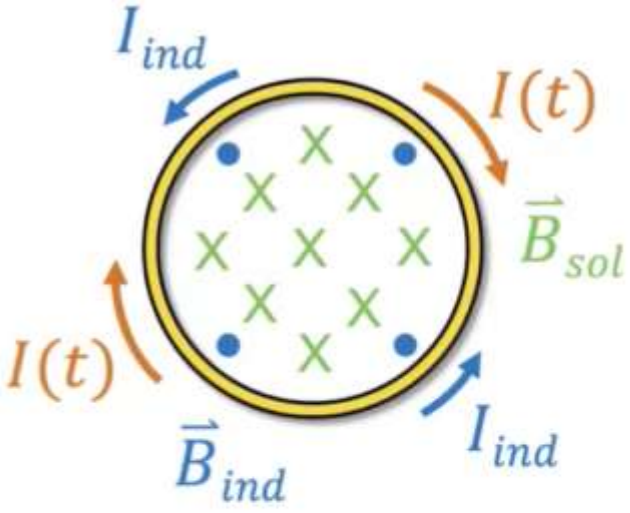
Inductance Defined

- Solenoids resist changes in current.



Inductance Defined

- Solenoids resist changes in current.
- As current increases, the solenoids magnetic field will increase.
- This will induce an emf that produces a current in a direction that opposes that change in flux.
- **<u>Inductance</u>:** The tendency of a conductor to resist a change in current.
- Units of Inductance: Henry



$$\frac{Tm^2}{A} = H \qquad L = \frac{N\Phi}{I}$$

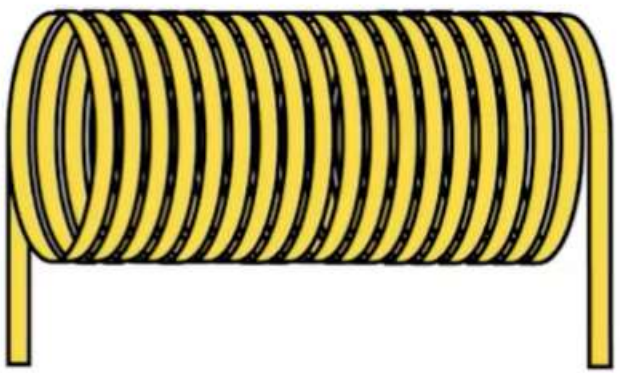
$$\mathrm{H} = \frac{\mathrm{kg{\cdot}m^2}}{\mathrm{s^2{\cdot}A^2}} = \frac{\mathrm{N{\cdot}m}}{\mathrm{A^2}} = \frac{\mathrm{J}}{\mathrm{A^2}} = \frac{\mathrm{kg{\cdot}m^2}}{\mathrm{C^2}} = \frac{\mathrm{s^2}}{\mathrm{F}}$$
$$= \frac{\mathrm{T{\cdot}m^2}}{\mathrm{A}} = \frac{\mathrm{Wb}}{\mathrm{A}} = \frac{\mathrm{V{\cdot}s}}{\mathrm{A}} = \frac{\Omega}{\mathrm{rad{\cdot}Hz}} = \frac{\Omega{\cdot}\mathrm{s}}{\mathrm{rad}}$$

where: H = henry, kg = kilogram, m = metre, s = second, A = ampere, N = newton, C = coulomb, J = joule, T = tesla, Wb = weber, V = volt, F = farad, Ω = ohm, Hz = hertz, rad = radian

The Solenoid as an Inductor

- **<u>Inductor</u>:** An electronic circuit element that possesses inductance.

- We can define the inductance of a solenoid just by looking at its physical features.



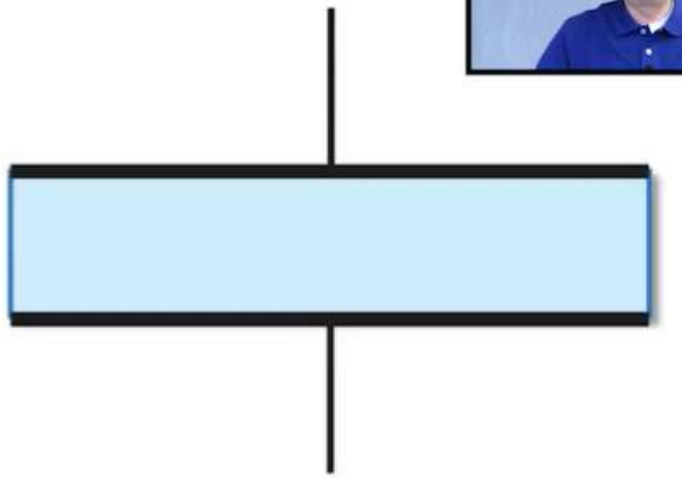
$$L = \frac{N\Phi}{I} \qquad B_{sol} = \mu_0 \frac{N}{l} I$$

$$L = \frac{N(B_{sol} A \cos\theta)}{I} = \frac{NA}{I} \cdot \mu_0 \frac{N}{l} I = \frac{\mu_0 N^2 A}{l} = L_{sol}$$

The Solenoid as an Inductor

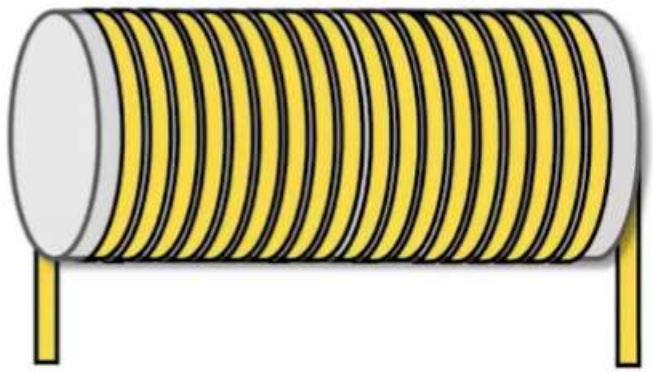
- A dielectric could increase the capacitance of a capacitor.

$$C = \frac{\kappa \epsilon_0 A}{d}$$



- A different material, other than air or a vacuum, in the core of a solenoid can increase its inductance.

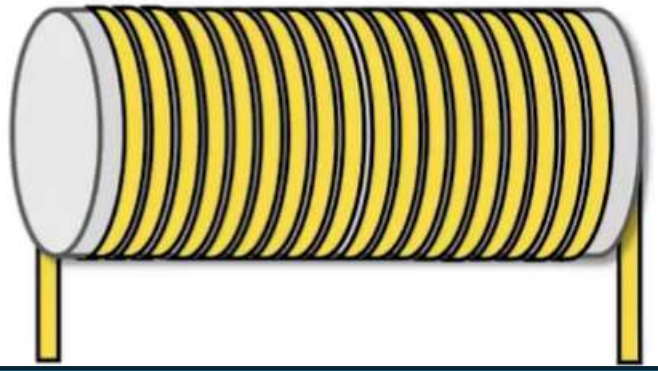
$$\boxed{L_{sol} = \frac{\mu_{core} N^2 A}{l}}$$



Calculating Inductance

A solenoid has 300 turns, a length of 5 cm long, and a diameter of 2.5mm. It also has an iron core which has a permeability of 2.51×10^{-4} Tm/A.

What is the overall inductance of this solenoid?



Takeaways

- Inductance is a property of a conductor to resist changes in current.
- Inductance can be explained by the arrangement of loops resulting in a change in magnetic flux when external factors are changed.
- A solenoid a a great example of an inductor.
- Like a capacitor, putting a different material inside a solenoid's core increases its inductance.
- Calculating the inductance of a solenoid can be done by focusing on physical characteristics independent of its electromagnetic behavior.

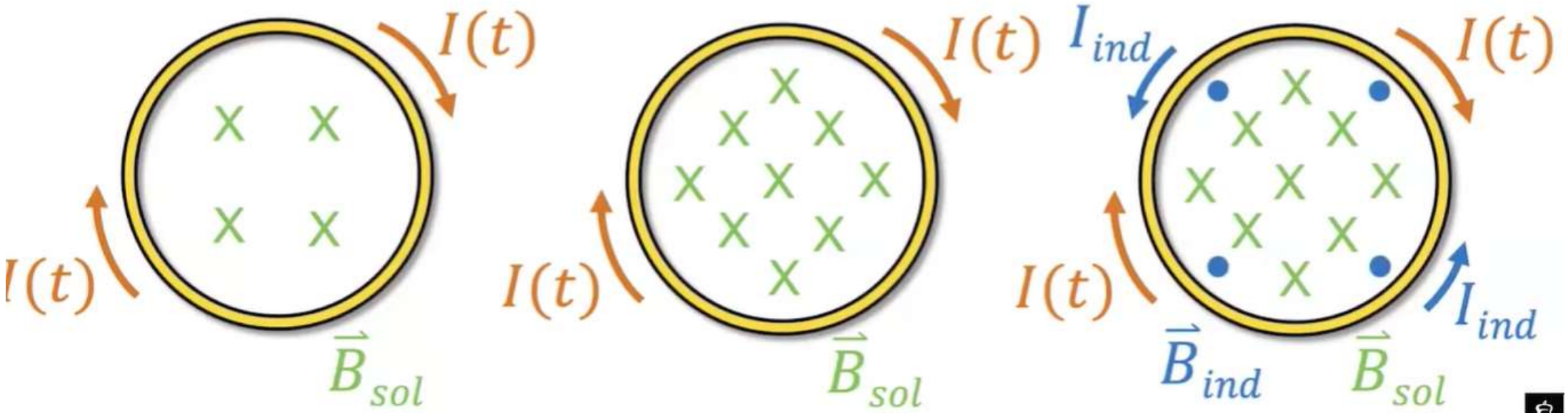
13.4.2

Self-Inductance of an inductor

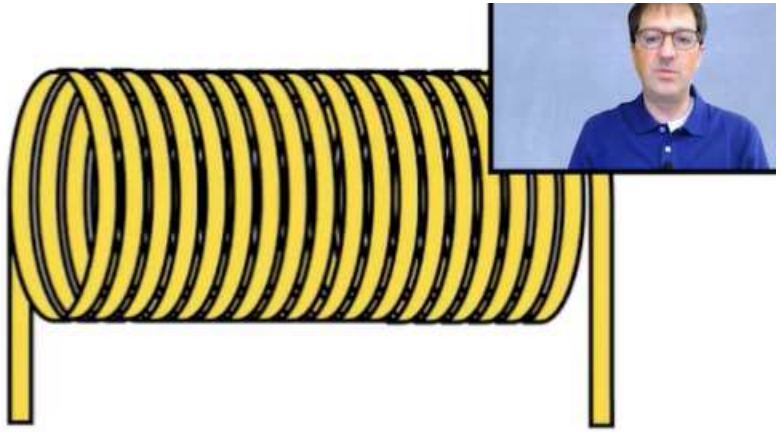
Deriving an equation for the induced emf when current changes

Resisting Changes in Current

- A solenoid with a current established in it produces a magnetic field through its core.
- As current changes, increasing in this example, the magnetic field increases in strength.
- This causes an induced current whose magnetic field opposes the change in the solenoid's flux.



Induced emf Quantified



- The outlined process explains the tendency of an inductor to oppose a change in current.
- Direction of this emf will always oppose a change in current.

$$\mathcal{E}_L = -N\frac{d\Phi}{dt} = -\frac{d(N\Phi)}{dt}$$

Mathematical Definition of Inductance:

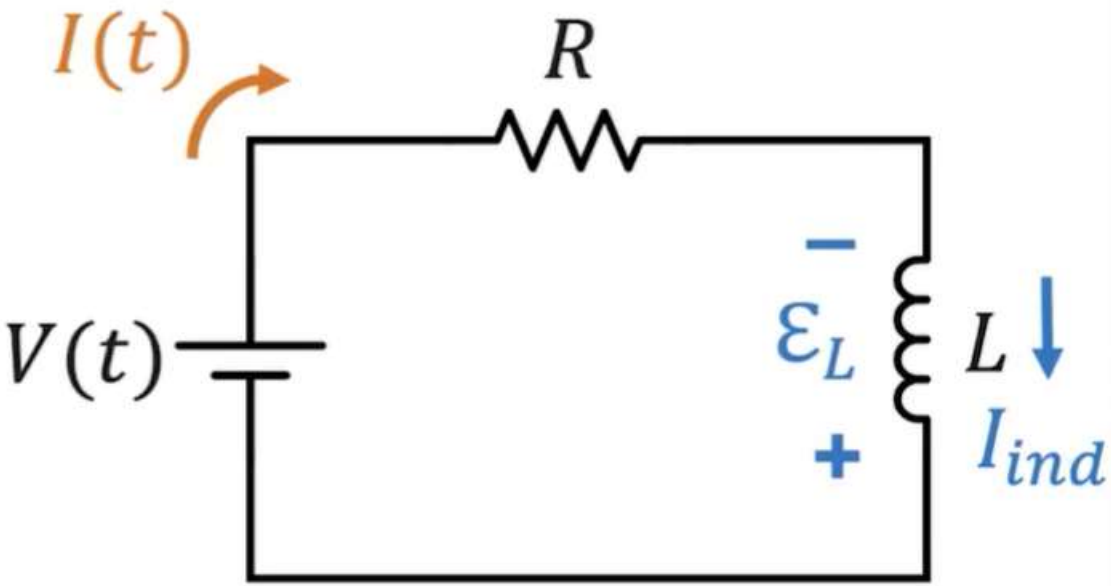
$$L = \frac{N\Phi}{I} \longrightarrow LI = N\Phi$$

$$\mathcal{E}_L = -\frac{d(LI)}{dt}$$

$$\boxed{\mathcal{E}_L = -L\frac{dI}{dt}}$$

Induced emf in a Circuit

- Inductors are used in circuits.
- As current changes, the inductor self-induces an emf that will oppose any change in current.
- If current increases, an emf is induced across the inductor.
- If current decreases, an emf is induced across the inductor.

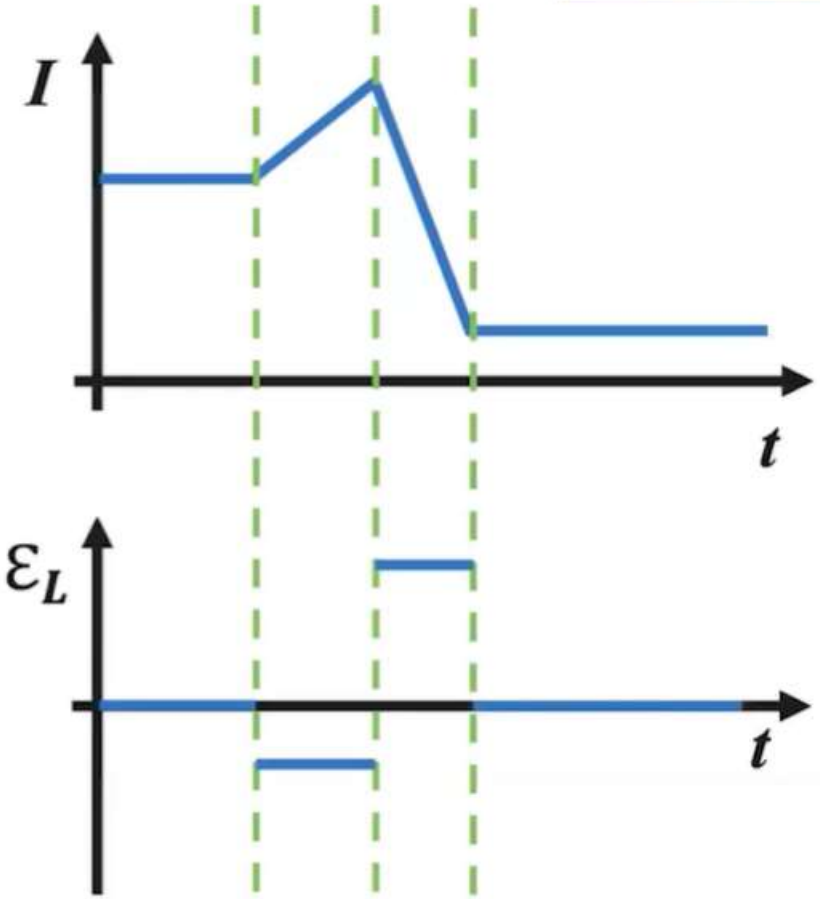


Graphing Induced emf

- The current through some inductor can be varied by its source and produces a current vs time graph as pictured to the right.

- Determine the relative shape of the graph representing the induced emf vs time.

$$\mathcal{E}_L = -L\frac{dI}{dt}$$



Takeaways

- An inductor will always induce an emf whenever current through its coils changes.
- This induced emf will always act in a direction to oppose the change or maintain its present state.
- It can be quantified by using the equation:

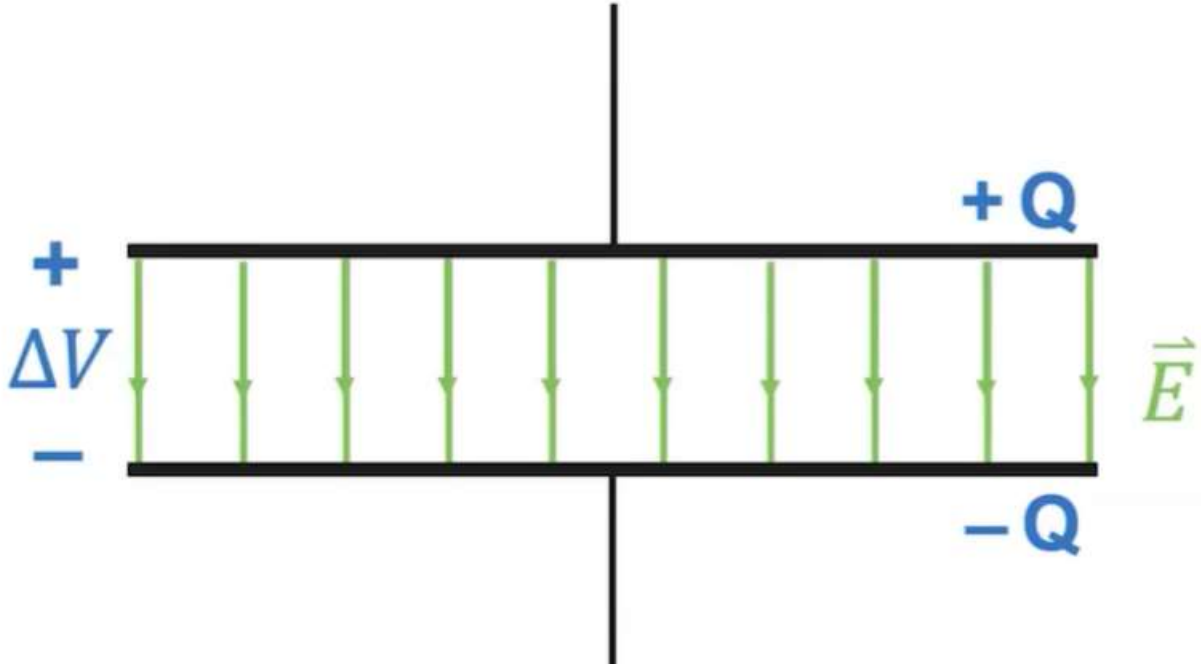
$$\varepsilon_L = -L\frac{dI}{dt}$$

13.4.3

Relating stored energy, inductance, and current to find the energy stored in the magnetic field of an inductor

Energy Storage Devices – The Capacitor

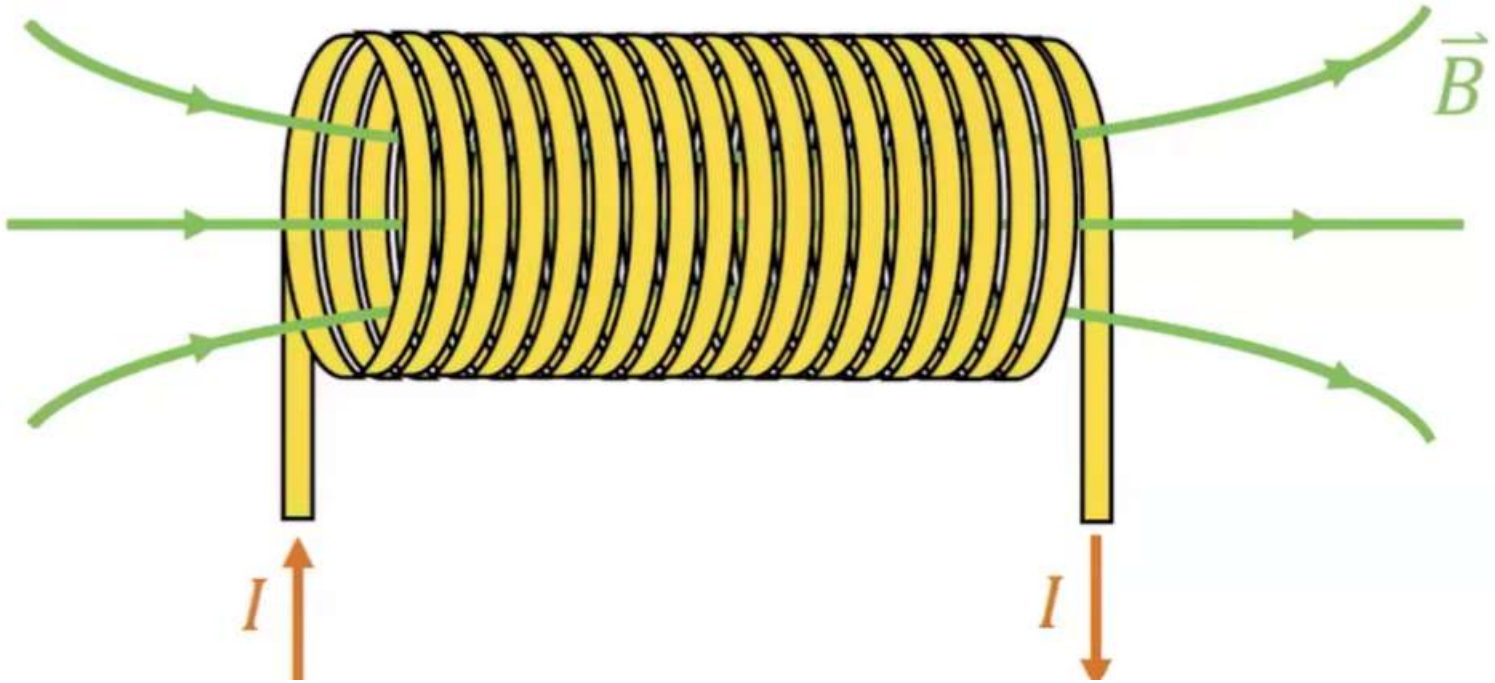
- When a capacitor is charged it possesses some potential difference ΔV and a
electric field $\vec{E}$ exists between the plates.
- Energy is stored in this electric field.



$$U_C = \frac{1}{2} Q \cdot \Delta V$$

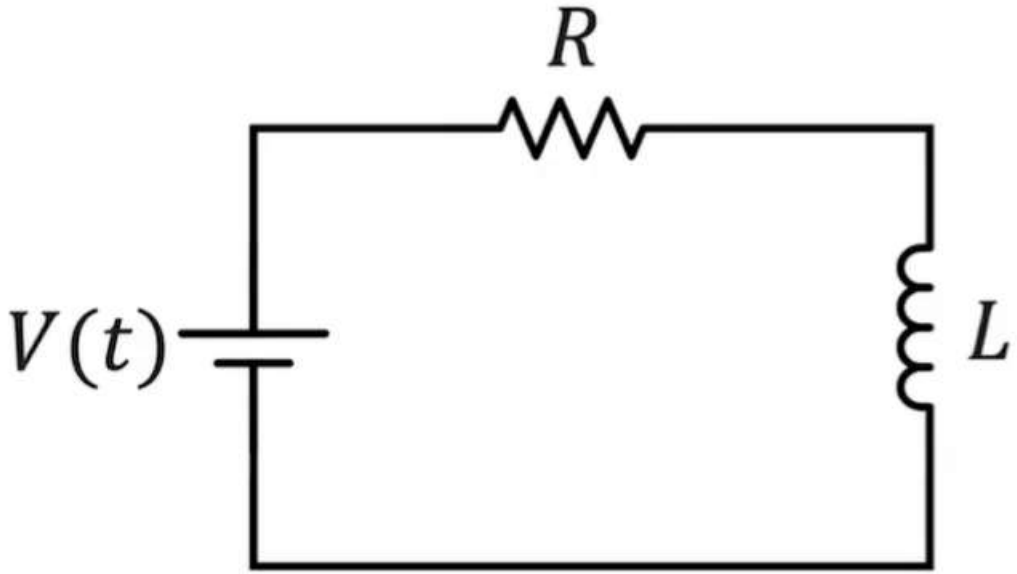
Energy Storage Devices – The Solenoid

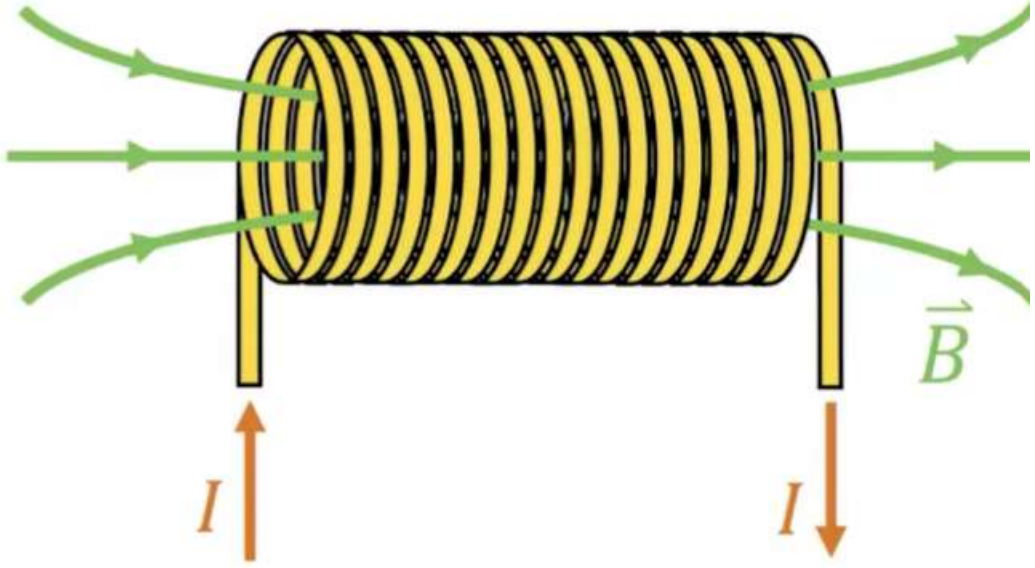
- When a current exists inside the coils of a solenoid, a magnetic field $\vec{B}$ exists through the core of the solenoid.
- Energy is stored in this magnetic field.



Energy Storage Devices – The Solenoid

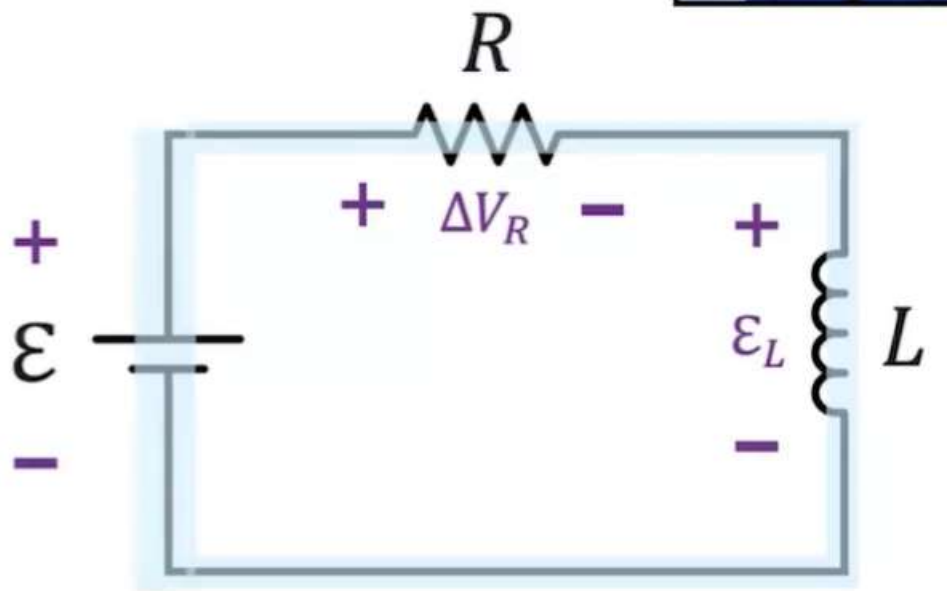
When we look at an application of an inductor in a simple circuit, we can quantity this energy stored in the field.





Determining Energy Stored in an Inductor

Kirchhoff's Loop Rule is an expression of conservation of energy applied to single loops of circuits.



$$\mathcal{E} - \Delta V_R - \mathcal{E}_L = 0$$

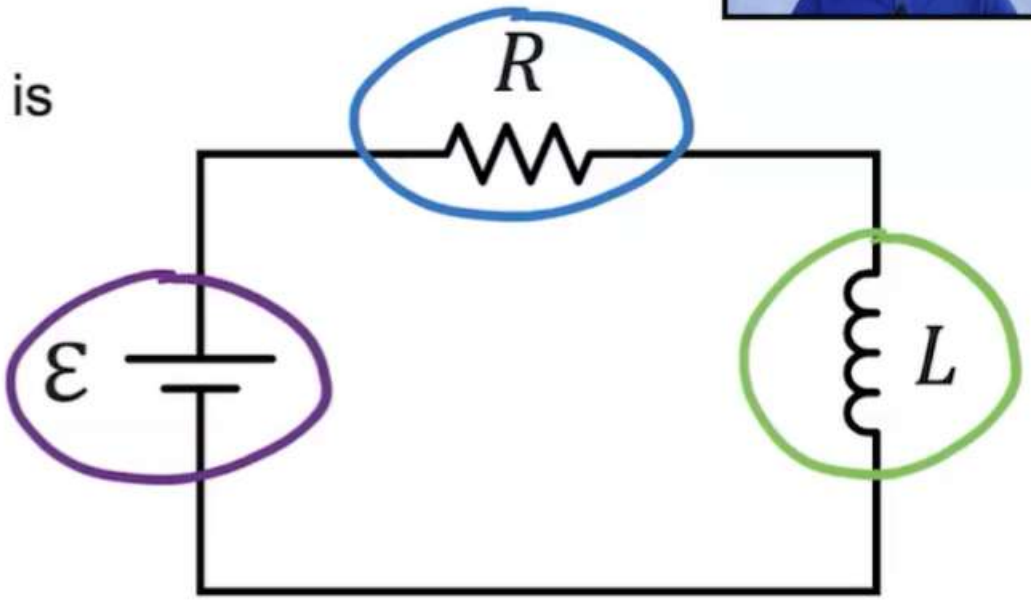
$$\mathcal{E} - IR - L\frac{dI}{dt} = 0$$

$$I \cdot \mathcal{E} - I \cdot IR - I \cdot L\frac{dI}{dt} = 0 \rightarrow \mathcal{E}I = I^2R + IL\frac{dI}{dt}$$

Determining Energy Stored in an Inductor

Each term describes a rate at which energy is supplied, dissipated or stored.

$$\mathcal{E}I = I^2R + IL\frac{dI}{dt}$$



- $\mathcal{E}I$ represents the power supplied to the circuit by the source emf.
- I^2R represents the rate at which energy is dissipated as thermal energy in the resistor.
- $IL\frac{dI}{dt}$ must, by conservation of energy, represent the rate at which energy is being stored or dissipated by the inductor.

Determining Energy Stored in an Inductor

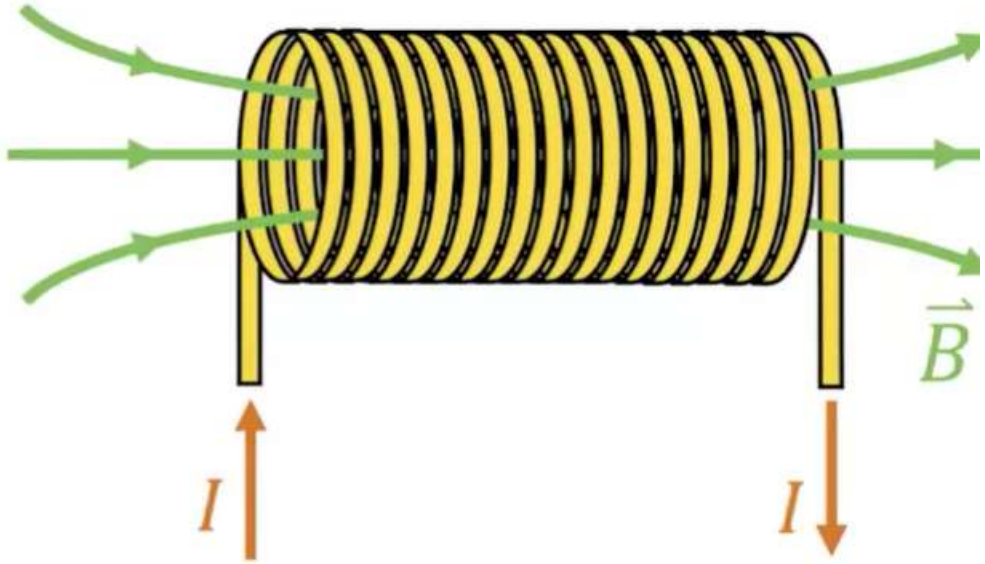
$$\frac{dU_L}{dt} = IL\frac{dI}{dt}$$

$$dU_L = ILdI$$

$$\int_0^{U_L} dU_L = \int_0^I ILdI$$

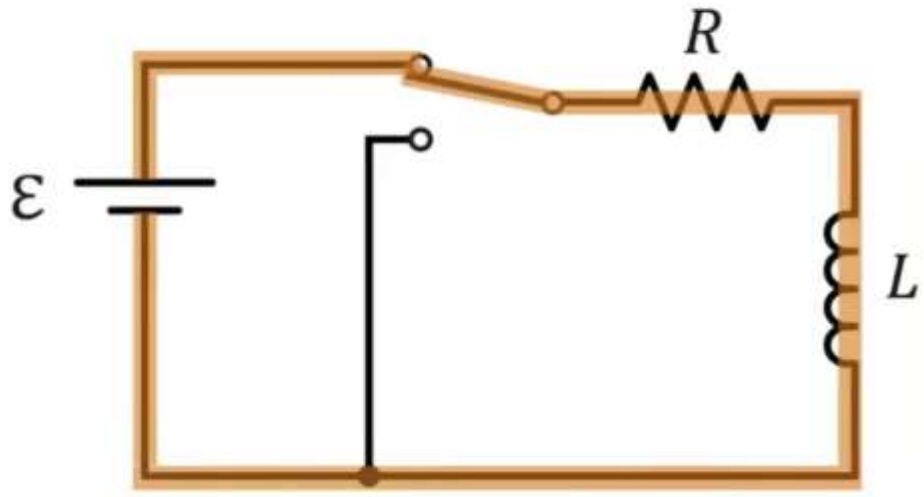
$$(U_L - 0) = \frac{L(I^2 - 0^2)}{2}$$

$$\boxed{U_L = \frac{1}{2}LI^2}$$



Electrical Energy in a RL Circuit

This energy can be dissipated through a resistor.



Electrical Energy in a RL Circuit

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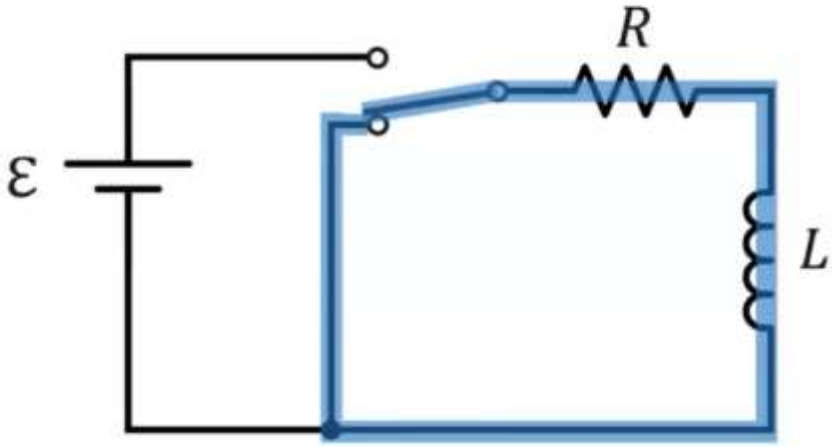
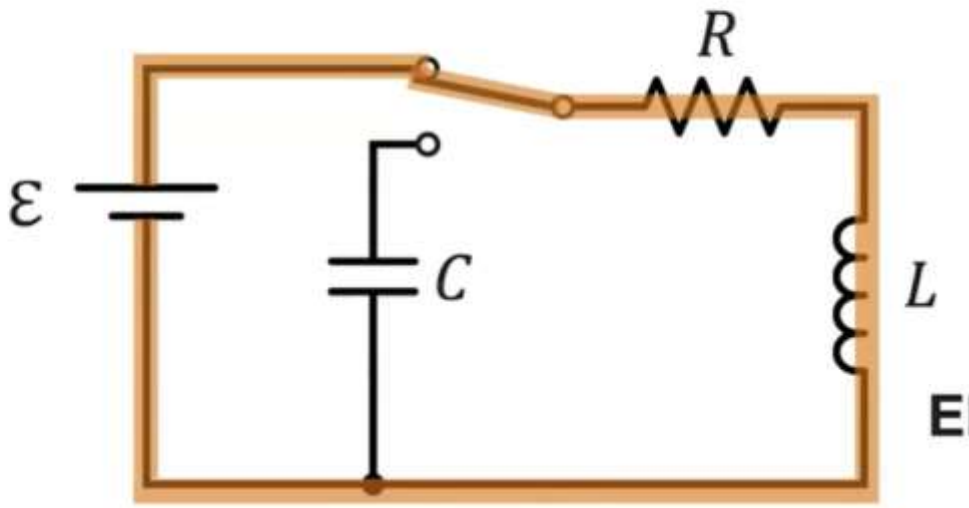


Electrical Energy in a RLC Circuit

This energy can also be used to charge a capacitor.



Electrical Energy in a RLC Circuit

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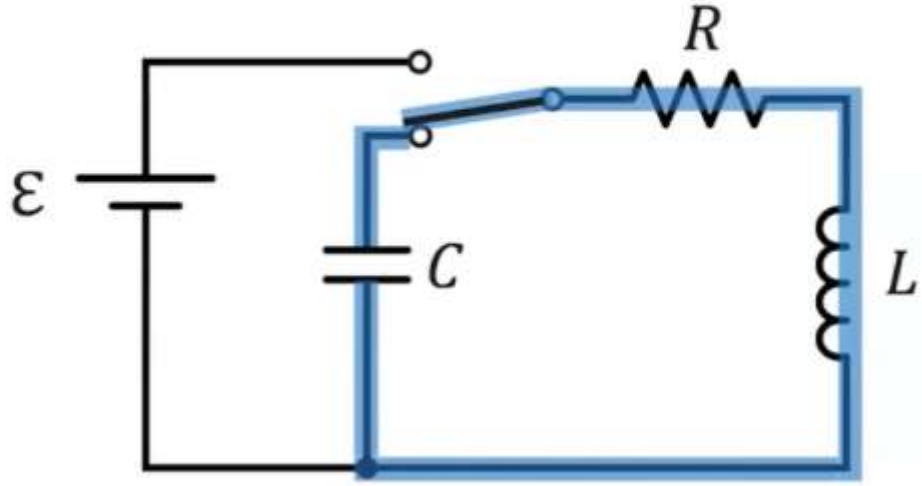

Takeaways

- Like a capacitor stores energy, so does an inductor.
- The equation to quantify the electrical energy stored in the inductor comes from a Kirchhoff's Loop Rule statement, ultimately an expression of conservation of energy.

$$U_L = \frac{1}{2} L I^2$$

- This energy can be dissipated in a resistor or used to charge a capacitor.